

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	28 september 2026
Team ID	SWTID-2026-4908
Project Name	Import Data using Transform Maps
Maximum Marks	4Marks

1. Functional Requirements (FR):

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Target Table Configuration	* Create custom table Employee Test (u_employee_test). * Configure 5 string columns: Employee ID, Employee Name, Email, Department, and Location.
FR-2	Data Ingestion & Staging	* Upload external Excel spreadsheet (Sample Spreadsheet.xlsx) via System Import Sets > Load Data. * Auto-generate staging import table u_employee_import to hold raw source data.
FR-3	Field Transformation &	* Create Employee

	Mapping	Transform Map connecting staging table to target table. * Execute Auto Map Matching Fields to map source headers directly to target attributes.
FR-4	Data Integrity & Duplicate Control	* Configure the Coalesce property on the Employee ID (u_employee_id) field map. * Ensure existing records update on matching key and new records insert when key is unique.
FR-5	Execution & Audit Logging	* Execute the Transform process manually or on file drop. * Generate Import Set logs to verify row count, successful inserts, updates, and error counts.

2. Non-Functional Requirements (NFR):

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The transform map process must leverage native ServiceNow UI options (like Auto Map Matching) so administrators can

		execute imports without writing complex background scripts.
NFR-2	Security	Access to import staging tables (u_employee_import) and target execution maps must be restricted via Access Control Lists (ACLs) to users with admin or import_admin roles
NFR-3	Reliability	The transform process must guarantee 100% data consistency—if an import fails mid-way, the Import Set log must identify exact corrupt rows without corrupting clean target records
NFR-4	Performance	The transform map must process bulk datasets efficiently, transforming standard batches of employee records within seconds without locking database threads.
NFR- 5	Availability	Import set tables and transform scripts must be available 24/7 in the ServiceNow instance,

		allowing automated imports whenever scheduled file drops occur.
NFR-6	Scalability	The design must support scaling from small sample files (5 rows) to enterprise legacy migrations (thousands of records) without changing core mapping logic.